PYTHON

Module 5: Output & Print

The print() Function

The `print()` function is the most fundamental tool in Python for communicating with the user. It sends data to the standard output device (usually your screen).

Functional Overview:

• Purpose: To display strings, numbers, and the results of calculations.

• Behavior: By default, every time you call `print()`, Python adds a newline character at the end, moving the cursor to the next line for the next instruction.

Handling Strings: Quotes

Textual data (Strings) must be wrapped in quotes so Python knows it is not looking at a command or variable name.

Flexibility:

- Double Quotes: `print("Hello CodeHive")`

- Single Quotes: `print('Hello CodeHive')`

Both are identical in Python. A SyntaxError occurs if quotes are missing, as Python will try to interpret the words as code logic rather than text.

Advanced Output: The 'end' Parameter

Sometimes you don't want a new line after every output. Python allows you to override the default behavior using the `end` parameter.

Scenario:

If you write `print("Hello", end=" ")` followed by `print("World")`, the output will be `Hello World` on a single line.

The value inside `end=""` can be a space, a comma, or even nothing at all, providing total control over your terminal's visual flow.

Printing Numeric Data

Unlike text, numbers do not require quotes. In fact, putting quotes around a number turns it into a string, which prevents mathematical operations.

Usage:

- Integers: `print(100)`

- Floats: `print(3.14)`

Mathematical Processing: You can perform arithmetic directly inside the parentheses: `print(10 * 5)` will output `50`.

Mixing Data Types

In real-world applications, you often need to combine text and numerical values. The simplest way to do this is by using a comma `,` to separate different data types.

Code Example:

```
print("The total is:", 42)
```

Advantage: When you use a comma, Python automatically inserts a space between the items, making the output readable without manual formatting.

Escape Characters

Inside your print() statement, you can use Special Characters to format text dynamically:

- `\n`: Inserts a new line within a single string.

- `\t`: Inserts a horizontal tab.

- `\\`: Allows you to print a literal backslash.

These tools allow for complex, multi-line output within a single statement.

Output Hierarchy & Summary

Mastering output is the first step toward building user interfaces. Remember these CodeHive tips:

1. Use clear, descriptive text labels when printing numbers.

2. Leverage the `end` parameter for progress bars or horizontal lists.

3. Double-check your quotes to avoid the common `SyntaxError` during your first steps in coding.